Software Test Plan (STP)

Event Management System — IEEE 829 structure

Project: Event Management System (EMS) — SE Mini Project
Version: 3.0
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Revision history

Version	Date	Author	Change summary	Approval
1.0	21 Oct 2026	Team	First issue (Sprint 1)	
2.0	04 Nov 2026	Team	Sprint 2 update	
3.0	13 Nov 2026	Team	Restructured to the course STP template (sections 1–15, 5.1 security validation); all tests re-run; load, restore and accessibility tests executed; 16 security cases; fail→fix demonstration	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Repository Owner	Amogh Herle		
Team members	Shravani Mallur, Arjun Bhat		

1. Introduction

Purpose: defines the test plan for EMS v1.0 — objectives, scope, strategy, environment, schedule, responsibilities, and the functional, non-functional and security test cases with their results.

Scope: all 20 functional requirements (FR-001a – FR-010b), all 7 non-functional requirements (NFR-001 – NFR-007) and the security requirements SR-001 – SR-012. Third-party services (PostgreSQL engine, SMTP provider, Vercel) are not re-tested.

References: SRS v3.0, SAD v3.0, SRSec, tests/*.test.ts, scripts/test_fr*.ts, .github/workflows/ci.yml, docs/reports/*.

Definitions: UT/IT/ST = unit/integration/system test; SEC = security validation case; NFT = non-functional test; RTM = requirements traceability matrix; “live” = executed against a running production build of main.

2. Test Items

Item	Requirement(s)	Primary module(s)
Account signup & email verification	FR-001a/b	app/api/auth/signup, verify
Login & password reset	FR-002a/b	app/api/auth/login, forgot-password, reset-password
RBAC, profile, admin user management	FR-003a/b/c	lib/auth.ts, app/api/auth/me, app/api/admin/users
Event creation & editing	FR-004a/b	app/api/events, lib/registration.ts
Event cancellation & notifications	FR-005a/b	app/api/events/[id], lib/registration.ts
Discovery, search, filter, categories	FR-006a/b/c	lib/eventFilters.ts, lib/categories.ts
Registration & duplicate prevention	FR-007	app/api/events/[id]/register
Capacity check & concurrency	FR-008	lib/registration.ts
Waitlist, self-cancellation, reminders	FR-009a/b	lib/registration.ts, app/api/reminders
Reporting, export, statistics	FR-010a/b	lib/export.ts, lib/stats.ts
Non-functional behaviour and security controls	NFR-001–007, SR-001–012	whole system

3. Features to be Tested

Every requirement in the SRS, mapped to the test cases that verify it (full matrix in §13).

Req ID	Feature	Test case(s)	Type
FR-001a	Account creation	FR001-UT_01, FR001-UT_02, FR001-ST_01, FR001-ST_02	Functional
FR-001b	Email verification	FR001-IT_01, FR001-IT_02, FR001-ST_01	Functional
FR-002a	Login	FR002-UT_01, FR002-UT_02, FR002-ST_01, FR003-ST_02	Functional
FR-002b	Password reset	FR002-IT_01, FR002-IT_02, FR002-ST_01, FR002-ST_02	Functional

FR-003a	Role-based access control	FR003-UT_01, FR003-ST_01, FR004-ST_02	Functional
FR-003b	Profile update	FR003-UT_02	Functional
FR-003c	Admin manages user accounts	FR003-IT_01, FR003-IT_02, FR003-ST_02	Functional
FR-004a	Create event	FR004-UT_01, FR004-UT_02, FR004-IT_02, FR004-ST_01, FR004-ST_02	Functional
FR-004b	Edit event	FR004-IT_01, FR004-IT_02, FR004-ST_01	Functional
FR-005a	Cancel or delete event	FR005-UT_01, FR005-UT_02, FR005-ST_02, FR007-ST_02	Functional
FR-005b	Notify participants of change	FR005-IT_01, FR005-IT_02, FR005-ST_01, FR005-ST_02	Functional
FR-006a	View events	FR006-ST_02	Functional
FR-006b	Search and filter events	FR006-UT_01, FR006-UT_02, FR006-IT_02, FR006-ST_01, FR006-ST_02	Functional
FR-006c	Admin manages categories	FR006-IT_01	Functional
FR-007	Register for event	FR007-UT_01, FR007-UT_02, FR007-IT_01, FR007-IT_02, FR007-ST_01, FR007-ST_02	Functional
FR-008	Capacity check	FR007-IT_01, FR008-UT_01, FR008-UT_02, FR008-IT_01, FR008-IT_02, FR008-ST_01, FR008-ST_02	Functional
FR-009a	Waitlist with auto-promotion	FR008-ST_02, FR009-UT_01, FR009-IT_01, FR009-ST_01	Functional
FR-009b	Self-cancellation and notifications	FR009-UT_02, FR009-IT_02, FR009-ST_02	Functional
FR-010a	Counts, seat availability and export	FR007-IT_02, FR010-UT_01, FR010-UT_02, FR010-IT_01, FR010-IT_02, FR010-ST_02	Functional
FR-010b	Admin platform statistics	FR010-ST_01	Functional
NFR-001	Performance	NFT_01, NFT_02	Non-functional
NFR-002	Scalability and concurrency	NFT_03, NFT_04	Non-functional
NFR-003	Security - data protection	NFT_05	Non-functional
NFR-004	Security - access control	NFT_06	Non-functional
NFR-005	Availability and recovery	NFT_07	Non-functional
NFR-006	Usability and accessibility	NFT_08, NFT_11	Non-functional
NFR-007	Maintainability	NFT_09, NFT_10	Non-functional

3.1 Security features

Security req	Feature	Test case(s)
SR-001	Passwords shall be stored only as salted bcrypt/Argon2 hashes and never logged.	SEC_01

SR-002	Session cookies shall be HttpOnly, SameSite and Secure (production), HMAC-signed, and shall expire after 30 minutes of inactivity.	SEC_02, SEC_11
SR-003	The account shall lock after 5 consecutive failed login attempts.	SEC_03
SR-004	RBAC and least privilege shall be enforced on the server for every request; no client-supplied role claim shall be trusted.	SEC_04, SEC_13
SR-005	Object references shall be authorized server-side (ownership checks) to prevent IDOR.	SEC_05
SR-006	All client-server traffic shall use TLS 1.2 or higher.	SEC_06
SR-007	Sensitive data at rest shall be encrypted.	SEC_06
SR-008	All database access shall use parameterized queries / an ORM.	SEC_07
SR-009	User-supplied input shall be validated and output encoded to prevent XSS.	SEC_08, SEC_15
SR-010	State-changing requests shall be protected against CSRF.	SEC_09
SR-011	Security events and administrative actions shall be recorded in a durable audit log (12 months) without secrets or plain-text passwords.	SEC_10
SR-012	Verification and reset tokens shall be random, single-use, time-limited and stored hashed; login errors shall not reveal whether an account exists.	SEC_12, SEC_14

4. Features not to be Tested

- PostgreSQL engine internals, the SMTP provider's delivery and the hosting platform's internals (assumed correct; EMS's own use of them is tested).
- TLS configuration and storage encryption of the hosting platform — not independently scannable locally (SEC_06 recorded “Not run”).
- Cross-browser rendering on Safari and Edge, and the first-time-user usability study (needs real participants/devices) — recorded as not executed in NFT_08.
- Native mobile apps, payments and external calendar integration (out of scope in the SRS).
- Production uptime (99.5%) — needs a month of monitoring.

5. Test Approach / Strategy

Levels

Level	ID prefix	What it exercises	Tooling
Unit	UT_	A single function/validation rule in isolation	node:test (tsx), in-memory/mocked Prisma where noted
Integration	IT_	A route + business logic + real PostgreSQL	node:test against Postgres (CI service container)
System	ST_	A user-facing flow across requests/roles	node:test route handlers; live scripts against a running production build; manual UI walkthrough where noted
Acceptance	—	SRS acceptance criteria and exit criteria	This plan §14

Types

- Functional (60 FR cases, §5.2) · Regression (270 automated tests on every push/PR) · Performance/concurrency (§5.3) · Security (§5.1) · Accessibility (axe-core) · Recovery drill (pg_dump/pg_restore)

Entry criteria: stable build on main, test database available, test data seeded. **Exit criteria:** 100% of planned cases executed or explicitly recorded as not run with reason; all High-priority FRs pass; every

failed or partial case has a documented deviation (SAD 4.6).

5.1 Security Validation

Each case traces to a security requirement (SR-xxx) and SRSec section. Results are from the 13 Nov 2026 run.

ID	SRSec §	Security requirement	Test description	Result	Evidence
SEC_01	2.1	SR-001	Passwords are stored as bcrypt hashes, never plain text or in logs.	**Pass**	Live TC-4 (hash starts $2, differs from password); code review of signup/reset-password.
SEC_02	2.2	SR-002	Session cookie is HttpOnly, SameSite=Lax, Secure in production; HMAC verified in constant time; tampered or expired cookies rejected.	**Pass**	tests/session.test.ts (13 tests).
SEC_03	2.3	SR-003	Account locks after 5 consecutive failed logins.	**Fail**	Not in main. Uncommitted local work exists (login route + schema) but has no tests and breaks `tsc` (Next route file exports a constant).
SEC_04	3.1/3.2	SR-004	A role without permission calling an endpoint directly gets 403 and no data.	**Pass**	api-routes (export 403), admin-categories (non-admins), admin-stats (RBAC), live TC-16/TC-21, fr003a TC-1.
SEC_05	3.3	SR-005	Organizer A cannot edit, cancel or export Organizer B's event (IDOR).	**Pass**	event-edit (403 non-owner), event-notifications (unauthorized cancel), api-routes (export / registrations 403 other organizer).
SEC_06	4.1/4.2	SR-006, SR-007	All traffic over TLS 1.2+; data encrypted at rest.	**Not run**	Provided by the hosting platform; no TLS scan or storage audit was possible locally. Secure cookie attribute in production is tested (SEC_02).
SEC_07	5.1	SR-008	SQL-injection payloads in search/filter/event fields have no effect.	**Pass**	event-search (special characters plain text; LIKE wildcards escaped); no raw SQL in lib/ or app/api/.
SEC_08	5.2	SR-009	Script payloads are escaped in notification email and page output.	**Pass**	event-notifications ('escape HTML'), email-templates tests.
SEC_09	5.3	SR-010	A cross-site request cannot perform a state-changing action with the victim's session.	**Partial**	SameSite=Lax only; no anti-CSRF token.
SEC_10	6.1/6.2	SR-011	Security events and admin actions are written to a durable audit log without secrets.	**Fail**	Not in main; only console.error. AuditLog implementation exists in uncommitted local work, untested.
SEC_11	2.2	SR-002	A session idle for 30 minutes is rejected.	**Fail**	lib/session.ts uses a fixed 7-day absolute TTL; no inactivity timeout (NFR-004 not met).
SEC_12	2.1	SR-012	Verification and reset tokens are single-use, expiring and stored hashed.	**Pass**	Live TC-8..TC-10; fr002b TC-5..TC-7; tokens stored as sha256 hash.

SEC_13	3.1	SR-004	An unauthenticated caller cannot act as Organizer or Participant by sending an x-user-role header or event_session_role cookie.	**Fail**	lib/auth.ts getAuthUserFromRequest falls back to the demo role claim when there is no session (fr003a TC-4 confirms an Organizer action succeeds). requireAdmin is not affected (real session only).
SEC_14	2.1	SR-012	Login and forgot-password do not reveal whether an email is registered.	**Pass**	fr002a TC-3 (same error), fr002b TC-2 (same message).
SEC_15	5.2	SR-009	CSV export neutralizes spreadsheet formula-injection payloads.	**Pass**	tests/registration.test.ts (formula-injection case).
SEC_16	3.1	SR-004	A suspended user's still-valid session stops working on the next request.	**Pass**	Live TC-17; fr003a TC-3.

Security summary: 10 Pass · 1 Partial · 4 Fail · 1 Not run (of 16). The failed/partial cases are tracked as deviations D3–D5 and D8–D9 in SAD 4.6.

5.2 Functional Test Cases

60 cases (6 per FR: 2 unit, 2 integration, 2 system). “Run” = **A** automated/live re-run on 13 Nov 2026, **M** manual UI walkthrough (4 Nov; not re-run). Full IDs have the form `FR001-UT_01`.

FR-001 — Account Registration & Email Verification

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR001-UT_01	Signup with valid details	201; account created unverified; login blocked until verified.	FR-001a	A	scripts/test_fr001_fr003bc.ts TC-1, TC-5	**Pass**
FR001-UT_02	Signup rejected for malformed email	400; no account created.	FR-001a	A	test_fr001_fr003bc TC-2	**Pass**
FR001-IT_01	Verification link activates the account	200; login succeeds afterwards.	FR-001b	A	test_fr001_fr003bc TC-6, TC-7	**Pass**
FR001-IT_02	Expired / used / invalid link rejected	400 expired / invalid; account stays inactive.	FR-001b	A	test_fr001_fr003bc TC-8, TC-9, TC-10	**Pass**
FR001-ST_01	End-to-end signup, verify, first login	User ends logged in.	FR-001a, FR-001b	A	test_fr001_fr003bc TC-1..TC-7	**Pass**
FR001-ST_02	Duplicate email rejected	409; no duplicate account.	FR-001a	A	test_fr001_fr003bc TC-3	**Pass**

FR-002 — Login & Password Reset

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR002-UT_01	Login with correct credentials	200 and session cookie set.	FR-002a	A	scripts/test_fr002a.ts TC-1	**Pass**
FR002-UT_02	Login with wrong password	401 generic error; no session.	FR-002a	A	test_fr002a TC-2, TC-3	**Pass**
FR002-IT_01	Forgot password sends reset link	200; identical response for unknown email.	FR-002b	A	scripts/test_fr002b.ts TC-1, TC-2	**Pass**
FR002-IT_02	Reset link updates password, old one invalid	200; old password 401, new 200.	FR-002b	A	test_fr002b TC-3, TC-4	**Pass**

FR002-ST_01	Forgot-password to login journey	Login with new password succeeds.	FR-002b, FR-002a	A	test_fr002b TC-1..TC-4	**Pass**
FR002-ST_02	Expired / reused reset link rejected	400 expired / invalid; password unchanged.	FR-002b	A	test_fr002b TC-5, TC-7	**Pass**

FR-003 — RBAC, Profile & Admin Account Management

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR003-UT_01	Participant blocked from Organizer function	403; no data returned.	FR-003a	A	scripts/test_fr003a.ts TC-1	**Pass**
FR003-UT_02	User updates own profile name	200; new name on next read.	FR-003b	A	test_fr001_fr003bc TC-11..TC-14	**Pass**
FR003-IT_01	Admin suspends a user	200; login 403 and live session invalid.	FR-003c	A	test_fr001_fr003bc TC-17, TC-18; fr003a TC-3	**Pass**
FR003-IT_02	Admin deletes a user	204; login then fails.	FR-003c	A	test_fr001_fr003bc TC-20, TC-21	**Pass**
FR003-ST_01	Each role sees only its permitted menu items	Only Admin sees Manage Users.	FR-003a	M	Manual UI walkthrough (4 Nov); API-level role gating re-run in fr003a TC-1/TC-2	**Pass**
FR003-ST_02	Suspended user gets clear error at login	403 'account suspended'; no session.	FR-003c, FR-002a	A	scripts/test_fr002a.ts TC-5	**Pass**

FR-004 — Event Creation & Update

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR004-UT_01	Event created with valid fields	201; event stored with all fields.	FR-004a	A	scripts/test_fr004a.ts UT_01	**Pass**
FR004-UT_02	Creation fails when a required field is missing	400 'Required field missing'.	FR-004a	A	test_fr004a UT_02; tests/validation.test.ts	**Pass**
FR004-IT_01	Edited details saved and listed immediately	PUT 200; GET shows new details.	FR-004b	A	tests/event-edit.test.ts (updated details returned by GET)	**Pass**
FR004-IT_02	Capacity accepts only a positive integer	400 'capacity must be a positive integer'.	FR-004a, FR-004b	A	test_fr004a IT_02; tests/validation.test.ts; event-edit non-positive capacity	**Pass**
FR004-ST_01	Organizer creates then edits; participant sees update	No stale data.	FR-004a, FR-004b	A	tests/event-edit.test.ts (saves all edited fields, no duplicate)	**Pass**
FR004-ST_02	Participant cannot create an event	403.	FR-004a, FR-003a	A	test_fr004a ST_02	**Pass**

FR-005 — Event Cancellation & Notifications

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR005-UT_01	Cancelling marks the event cancelled	Status CANCELLED.	FR-005a	A	scripts/test_fr005a.ts TC-2	**Pass**
FR005-UT_02	Cancelled event rejects registrations	400 event cancelled.	FR-005a	A	tests/api-routes.test.ts (register returns 400 when event is cancelled)	**Pass**

FR005-IT_01	Cancelling notifies every registrant	One notification per participant.	FR-005b	A	tests/event-notifications.test.ts (notifies confirmed and waitlisted)	**Pass**
FR005-IT_02	Notification content names the event and changes	Email shows event and change.	FR-005b	A	event-notifications (changes and escape HTML; renamed event)	**Pass**
FR005-ST_01	Registered participant receives the cancellation email	Email captured.	FR-005b	A	event-notifications; mailer jsonTransport capture	**Pass**
FR005-ST_02	Cancelling an event with no registrants	Success; zero notifications.	FR-005a, FR-005b	A	event-notifications (zero participants); fr005a TC-2	**Pass**

FR-006 — Discovery, Search, Filtering & Categories

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR006-UT_01	Search by exact event name	Only the matching event returned.	FR-006b	A	tests/event-search.test.ts (full name)	**Pass**
FR006-UT_02	Filter by category	Only that category.	FR-006b	A	event-search (category filter)	**Pass**
FR006-IT_01	Admin-added category appears in the creation form	Category selectable.	FR-006c	A	tests/admin-categories.test.ts (add; event creation links category)	**Pass**
FR006-IT_02	Filter by date	Only events on that date.	FR-006b	A	event-search (date filter)	**Pass**
FR006-ST_01	Search + category + date together	Intersection only.	FR-006b	A	event-search (combine filters)	**Pass**
FR006-ST_02	Cancelled events excluded from listing and search	Not shown.	FR-006a, FR-006b	A	event-search (keep excluding cancelled); fr005a TC-3	**Pass**

FR-007 — Event Registration & Duplicate Prevention

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR007-UT_01	Valid registration creates a correct record	201; user/event/status stored.	FR-007	A	tests/registration.test.ts FR-007 / UT_01	**Pass**
FR007-UT_02	Duplicate registration rejected	409 already registered.	FR-007	A	registration.test FR-007 / UT_02; live 100-duplicate run (1 row)	**Pass**
FR007-IT_01	Registration decrements the seat count by one	Remaining seats N-1.	FR-007, FR-008	A	registration.test FR-008 / UT_02	**Pass**
FR007-IT_02	New registration appears in organizer's registrant list	Listed.	FR-007, FR-010a	A	tests/api-routes.test.ts (registrations list; organizer totals)	**Pass**
FR007-ST_01	Register, see confirmation, organizer sees entry	End to end.	FR-007	A	api-routes (creates registration and GET lists it)	**Pass**
FR007-ST_02	Cancelled or non-existent event rejected	400 / 404.	FR-007, FR-005a	A	registration.test FR-007 / ST_02	**Pass**

FR-008 — Seat Capacity Check & Concurrency

Test ID	Description	Expected result	Req	Run	Verified by	Result

FR008-UT_01	Registration rejected when zero seats remain	400 event full.	FR-008	A	registration.test FR-008 / UT_01	**Pass**
FR008-UT_02	Registration succeeds when a seat is available	Seat count to 0.	FR-008	A	registration.test FR-008 / UT_02	**Pass**
FR008-IT_01	Only one of two simultaneous requests for last seat wins	Exactly one confirmed.	FR-008	A	registration.test FR-008 / IT_01	**Pass**
FR008-IT_02	Capacity never negative under 10 concurrent requests	Final count 0.	FR-008	A	registration.test FR-008 / IT_02	**Pass**
FR008-ST_01	100 concurrent requests for a 1-seat event	Exactly 1 confirmed, 99 rejected.	FR-008	A	registration.test FR-008 / ST_01; live production-build run: 1 x 201, 99 x 400, 1 DB row	**Pass**
FR008-ST_02	Participant sees 'Event Full' and cannot register	Full message / disabled button.	FR-008, FR-009a	M	Manual UI walkthrough (4 Nov); API 'event full' re-run in api-routes tests	**Pass**

FR-009 — Waitlist, Self-Cancellation & Notifications

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR009-UT_01	Joining waitlist records queue position	Position returned.	FR-009a	A	registration.test FR-009a / UT_01	**Pass**
FR009-UT_02	Self-cancel frees the seat	Status cancelled; seats +1.	FR-009b	A	registration.test FR-009b / UT_02	**Pass**
FR009-IT_01	Cancellation promotes the first waitlisted user	Promoted and notified.	FR-009a	A	registration.test FR-009a / IT_01	**Pass**
FR009-IT_02	Confirmation email sent after registration	Email captured.	FR-009b	A	registration.test FR-009b / IT_02	**Pass**
FR009-ST_01	Waitlisted user auto-promoted and notified	Status confirmed + email.	FR-009a	A	registration.test FR-009a / ST_01	**Pass**
FR009-ST_02	Reminder sent before event, not re-sent	One reminder per registrant.	FR-009b	A	registration.test FR-009b / ST_02; reminders fallback tests	**Pass**

FR-010 — Reporting, Export & Statistics

Test ID	Description	Expected result	Req	Run	Verified by	Result
FR010-UT_01	Displayed registered count equals active registrations	Exact match.	FR-010a	A	registration.test FR-010a / UT_01	**Pass**
FR010-UT_02	Available seats = capacity minus active registrations	Exact match.	FR-010a	A	registration.test FR-010a / UT_02	**Pass**
FR010-IT_01	CSV export matches the on-screen list	Same rows.	FR-010a	A	registration.test FR-010a / IT_01; api-routes export csv	**Pass**
FR010-IT_02	PDF export matches the on-screen list	Same rows.	FR-010a	A	registration.test FR-010a / IT_02; api-routes export pdf	**Pass**
FR010-ST_01	Admin statistics equal database totals	50 events / 300 users / 900 registrations.	FR-010b	A	tests/admin-stats.test.ts FR-010b / ST_01	**Pass**
FR010-ST_02	Organizer sees accurate per-event info for several events	No cross-event mixing.	FR-010a	A	registration.test FR-010a / ST_02	**Pass**

5.3 Non-Functional Test Cases

ID	NFR	Description	Result	Evidence

NFT_01	NFR-001	CSV and PDF export of 5,000 registrations completes within 10 s.	**Pass**	Live run on main production build: CSV 216 ms, PDF 226 ms; also tests/registration.test.ts.
NFT_02	NFR-001	At 500 concurrent users: event list/search/filter p95 <= 2 s and registration p95 <= 3 s (dataset: 10,000 users, 1,000 events, 100,000 registrations).	**Fail**	Local single-machine run: list/search p95 8.3 s (p50 5.3 s); 500 simultaneous registrations p95 121.7 s, 0 errors. Targets NOT met; app, DB and load generator shared one laptop, so absolute numbers are pessimistic, but registrations appear to be serialized.
NFT_03	NFR-002	100 simultaneous requests for a 1-seat event confirm exactly one; 100 duplicate requests from one user create one record.	**Pass**	Live run at full scale: 1 x 201 + 99 x 400, 1 confirmed row; duplicates: 1 x 201 + 99 x 409, 1 row. Also registration.test.ts.
NFT_04	NFR-002	Waitlist entries are promoted in strict FIFO order under concurrent cancellations.	**Pass**	tests/registration.test.ts FR-009a / NFR-002.
NFT_05	NFR-003	Passwords hashed; tokens single-use, expiring and hashed; no plain-text secrets.	**Pass**	See SEC_01, SEC_12 (live TC-4, TC-8..TC-10; fr002b TC-5..TC-7).
NFT_06	NFR-004	RBAC, session timeout, account lockout, audit log.	**Partial**	RBAC pass (SEC_04/05), lockout and audit log not in main (SEC_03, SEC_10), 30-min idle timeout not implemented (SEC_11), demo-role header accepted (SEC_13).
NFT_07	NFR-005	Restore drill from backup completes within 4 h with <= 24 h data loss; uptime >= 99.5%.	**Partial**	Local drill (10,000 users / 1,000 events / 100,405 registrations): pg_dump 0.38 s, pg_restore into a fresh DB 1.0 s, row counts identical. Daily scheduled backup and monthly uptime monitoring are not configured, so RPO and 99.5% uptime are not demonstrated.
NFT_08	NFR-006	WCAG 2.1 A/AA automated scan (axe-core 4.10.2) and a first-time-user study (>= 90% in 3 min).	**Partial**	axe-core: 0 violations on 9 pages at 1280 px and at 320 px (home, login, signup, forgot-password, events, event detail, create event, my registrations, admin). The usability study was not run (needs human participants).
NFT_09	NFR-007	Automated line coverage of core business logic >= 80%.	**Pass**	270 tests; 92.29% overall lines; lib/registration.ts 92.77%, lib/auth.ts 92.27%, lib/mailer.ts 100%, lib/emailTemplates.ts 100% (docs/reports/coverage-report.txt).
NFT_10	NFR-007	A forced application error appears in a central log with timestamp and severity.	**Fail**	Main only calls console.error. A durable AuditLog exists only in uncommitted local work (PR #64 for NFR-007 is open, conflicting).
NFT_11	NFR-006	No horizontal scroll at 320 px on key pages.	**Fail**	7 of 9 pages overflow horizontally at 320 px on main; fixed by open PR #63 (feature/nfr-006-accessibility, not yet reviewed or merged).

6. Test Environment

Element	Detail
Runtime	Node.js 22 (CI: Node 20), TypeScript, Next.js 14.2 App Router, Prisma 5.22
Database	PostgreSQL 16 (Docker) — real database, never a mock for concurrency cases; separate throw-away databases for the test, load and restore runs
Mail	nodemailer jsonTransport in test (no real send)
CI	GitHub Actions (.github/workflows/ci.yml): typecheck, npm test, coverage, production build
Load/restore host	One Linux laptop running app, PostgreSQL and the load generator together (pessimistic for latency)

Tools	tsx --test, Node fetch load script, axe-core 4.10.2 + puppeteer-core with Chrome, pg_dump/pg_restore, GitHub Issues/Projects
Test data	Seeded: 10,000 users, 1,000 events, 100,000 registrations (load); generated test users per run

7. Test Schedule

Milestone	Date
Sprint 1 (test case design and first execution)	19 Oct – 25 Oct 2026
Sprint 2 (automated suite, CI, Test Plan v2)	26 Oct – 31 Oct 2026; plan issued 4 Nov
Sprint 3 (NFR work and final re-execution)	4 Nov – 10 Nov 2026
Full re-run incl. load, recovery, accessibility, security	13 Nov 2026

8. Test Deliverables

- This Test Plan and Documents/TEST-CASE_DOCUMENT.pdf (Sprint 1 execution log)
- Automated tests: tests/*.test.ts (270) and scripts/test_fr*.ts (74 live checks)
- Reports (docs/reports/): test-run.txt, coverage-report.txt, live-api-test-run.txt, fault-injection/ (fail→fix demonstration)
- Documents/TRACEABILITY_MATRICES.pdf (SRS→Test and Security→Test lists)
- Defect/gap list: SAD 4.6 and GitHub issues

9. Roles and Responsibilities

Role	Name	Responsibility
QA / Test Lead	Amogh Herle (PES1UG24CS653)	Plan, execute and report; owns FR-001–003 and security tests
Test Engineer / Developer	Shravani Mallur (PES1UG24CS621)	Events and notifications tests (FR-004–006), accessibility (NFR-006)
Test Engineer / Developer	Arjun Bhat (PES1UG24CS655)	Registration, concurrency and reporting tests (FR-007–010)
Product Owner / Approver	Repository owner and course coordinator	Review PRs, approve results and sign off

10. Risks and Mitigation

Risk	Mitigation
Concurrency defects invisible to mocks	Run concurrency cases against real PostgreSQL; verified at 100 concurrent requests
Local load test not representative of production	Report numbers with caveats; repeat on a staging deployment
Unreviewed merges break main	CI on every PR; reverts #52–#54; require review before merge
Uncommitted local changes diverge from main	Results are for main; local WIP flagged separately (SEC_03, SEC_10)
No human participants for usability study	Recorded as not executed; schedule a session

11. Assumptions & Dependencies

- PostgreSQL 16 and Node are available to run the suite; CI service container is healthy.
- Test accounts and data are generated per run; no production data is used.
- SMTP is replaced by jsonTransport in tests.
- Hosting platform provides TLS and storage encryption as documented.

12. Suspension & Resumption Criteria

Suspend if the CI pipeline or Postgres service cannot start (blocks all IT_/ST_ cases), or the build is too unstable (blocks >30% of cases). **Resume** when the blocking defect or environment issue is fixed; previously passing cases are re-run only if the fix touched code they exercise.

13. Test Case Management & Traceability

RTM maps every SRS requirement to its test cases (forward) and every test case group back to its requirement (reverse). The separate printable lists are in Documents/TRACEABILITY_MATRICES.pdf.

13.1 SRS → Test (forward)

Req ID	Requirement	Test case(s)	Overall result
FR-001a	Account creation	FR001-UT_01, FR001-UT_02, FR001-ST_01, FR001-ST_02	**Pass**
FR-001b	Email verification	FR001-IT_01, FR001-IT_02, FR001-ST_01	**Pass**
FR-002a	Login	FR002-UT_01, FR002-UT_02, FR002-ST_01, FR003-ST_02	**Pass**
FR-002b	Password reset	FR002-IT_01, FR002-IT_02, FR002-ST_01, FR002-ST_02	**Pass**
FR-003a	Role-based access control	FR003-UT_01, FR003-ST_01, FR004-ST_02	**Pass**
FR-003b	Profile update	FR003-UT_02	**Pass**
FR-003c	Admin manages user accounts	FR003-IT_01, FR003-IT_02, FR003-ST_02	**Pass**
FR-004a	Create event	FR004-UT_01, FR004-UT_02, FR004-IT_02, FR004-ST_01, FR004-ST_02	**Pass**
FR-004b	Edit event	FR004-IT_01, FR004-IT_02, FR004-ST_01	**Pass**
FR-005a	Cancel or delete event	FR005-UT_01, FR005-UT_02, FR005-ST_02, FR007-ST_02	**Pass**
FR-005b	Notify participants of change	FR005-IT_01, FR005-IT_02, FR005-ST_01, FR005-ST_02	**Pass**
FR-006a	View events	FR006-ST_02	**Pass**
FR-006b	Search and filter events	FR006-UT_01, FR006-UT_02, FR006-IT_02, FR006-ST_01, FR006-ST_02	**Pass**
FR-006c	Admin manages categories	FR006-IT_01	**Pass**
FR-007	Register for event	FR007-UT_01, FR007-UT_02, FR007-IT_01, FR007-IT_02, FR007-ST_01, FR007-ST_02	**Pass**

FR-008	Capacity check	FR007-IT_01, FR008-UT_01, FR008-UT_02, FR008-IT_01, FR008-IT_02, FR008-ST_01, FR008-ST_02	**Pass**
FR-009a	Waitlist with auto-promotion	FR008-ST_02, FR009-UT_01, FR009-IT_01, FR009-ST_01	**Pass**
FR-009b	Self-cancellation and notifications	FR009-UT_02, FR009-IT_02, FR009-ST_02	**Pass**
FR-010a	Counts, seat availability and export	FR007-IT_02, FR010-UT_01, FR010-UT_02, FR010-IT_01, FR010-IT_02, FR010-ST_02	**Pass**
FR-010b	Admin platform statistics	FR010-ST_01	**Pass**
NFR-001	Event list and search/filter pages shall load within 2 s and a registration shall complete within 3 s at the 95th percentile with 500 concurrent users; a 5,000-record CSV/PDF export shall complete within 10 s.	NFT_01, NFT_02	**Fail**
NFR-002	The system shall support 10,000 users, 1,000 active events and 100,000 registrations without degradation and shall execute concurrent operations atomically (no overbooking, duplicate or lost update).	NFT_03, NFT_04	**Pass**
NFR-003	Passwords shall be stored only as salted hashes (bcrypt/scrypt/Argon2); verification/reset links shall be single-use, random and time-limited; all traffic shall use TLS 1.2+; sensitive data at rest shall be encrypted.	NFT_05, SEC_01, SEC_06, SEC_12	**Pass**
NFR-004	RBAC shall be enforced server-side on every request; sessions shall expire after 30 minutes of inactivity; accounts shall lock after 5 consecutive failed logins; OWASP Top 10 protections shall apply; administrative actions shall be audit-logged for 12 months.	NFT_06, SEC_03, SEC_04, SEC_05, SEC_09, SEC_10, SEC_11, SEC_13	**Partial**
NFR-005	The system shall be available 99.5% of each month (excluding <= 4 h announced maintenance); data shall be backed up daily and restorable with RPO <= 24 h and RTO <= 4 h.	NFT_07	**Partial**
NFR-006	A first-time participant shall find an event and register within 3 minutes without training; the interface shall meet WCAG 2.1 AA, be responsive from 320 px upward and work on the latest two versions of Chrome, Firefox, Safari and Edge.	NFT_08, NFT_11	**Fail**
NFR-007	Application errors shall be logged centrally with timestamp and severity, and the codebase shall have at least 80% automated test coverage of core business logic.	NFT_09, NFT_10	**Fail**

13.2 Security requirement → Test

Security req	SRSec §	Test case(s)	Result
SR-001	2.1	SEC_01	**Met**
SR-002	2.2	SEC_02, SEC_11	**Partial**
SR-003	2.3	SEC_03	**Not met in main**
SR-004	3.1, 3.2	SEC_04, SEC_13	**Partial**
SR-005	3.3	SEC_05	**Met**
SR-006	4.1	SEC_06	**Met (hosting)**
SR-007	4.2	SEC_06	**Met (hosting)**

SR-008	5.1	SEC_07	**Met**
SR-009	5.2	SEC_08, SEC_15	**Met**
SR-010	5.3	SEC_09	**Partial**
SR-011	6.1, 6.2	SEC_10	**Not met in main**
SR-012	2.1	SEC_12, SEC_14	**Met**

14. Test Metrics & Reporting

Metric	Value (13 Nov 2026)
Automated tests (tests/*.test.ts, 15 files)	270 / 270 pass, 0 fail
Live API checks (scripts/test_fr*.ts, 6 scripts)	74 / 74 pass, 0 fail
Functional cases (§5.2)	60 / 60 Pass (58 verified by automated re-run today, 2 manual UI cases carried from 4 Nov)
Non-functional cases (§5.3)	5 Pass · 3 Partial · 3 Fail (of 11)
Security validation (§5.1)	10 Pass · 1 Partial · 4 Fail · 1 Not run (of 16)
Line coverage	92.29% overall; core modules ≥ 92% (NFR-007 target 80%)
Requirement coverage	20/20 FRs and 7/7 NFRs have at least one executed test case; 12/12 security requirements mapped
Fail → fix demonstration	3 injected faults: 250/270 pass (20 fail), after correction 270/270 (docs/reports/fault-injection/)

Open items (not silently marked Pass)

- **NFT_02** latency at 500 concurrent users fails (D2).
- **SEC_03, SEC_10, SEC_11, SEC_13** fail; **SEC_09** partial (D3–D5, D8).
- **NFT_08/NFT_11** usability study not run; 320 px overflow on main (D7).
- **NFT_07** daily backup and uptime monitoring not configured (D6); **SEC_06** not independently run (D9).

Reports: daily execution status via CI; final summary is this section and docs/reports/.

15. Approvals

Role	Name	Signature / Date
QA Lead	Amogh Herle	
Dev Lead	Shravani Mallur / Arjun Bhat	
Product Owner / Course Coordinator		